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Title of the article: *Dental pulp cell-derived powerful inducer of TNF- comprises pKR containing stress granule rich microvesicles*

The content extracted is as follows:

It is well known that dental pulp tissue can evoke some of the most severe acute inflammation observed in the human body.

We found that dental pulp cells secrete a factor that induces tumor necrosis factor-production from macrophages, and designated this factor, dental pulp cell-derived powerful inducer of TNF- (DpIt).

DpIt was induced in dental pulp cells and transported to recipient cells via microvesicles.

Treatment of dental pulp cells with a PKR inhibitor markedly suppressed DPIT activity, and weak interferon signals were constitutively activated inside the cells.

Proteomics analyses revealed that many stress granule-related proteins were present in supernatants from dental pulp cells as well as microvesicle marker proteins like GAPDH, -actin, HSPA8, HSPB1, HSPE1, and HSPD1.

Furthermore, giant molecule AHNAK and PKR were detected in microvesicles derived from dental pulp cells, and gene silencing of AHNAK in dental pulp cells led to reduced DPIT activity.

Once acute inflammation has been evoked in dental pulp tissue, the inflammation is rapidly up-regulated inside the tooth and the tissue usually undergoes complete necrosis within a few days¹.

We found that dental pulp cells, both immortalized cells² and primary cells, secrete a factor that strongly stimulates differentiated THP-1 (dTHP-1) cells to induce tumor necrosis factor (TNF)- at both the gene and protein levels (Fig. 1a: left, 1b: left), and designated this factor dental pulp cell-derived powerful inducer of TNF- (DPIT).

DPIT activity was seen in both immortalized dental pulp cells (DP-1) and primary dental pulp cells (PriDPC) and the activity was superior to that of a gram-negative bacterial endotoxin, lipopolysaccharide (LPS) when dTHP-1 cells were incubated for 2 hrs (Fig. 1a: right, 1b: right).

Culture supernatants from dental pulp cells also stimulated dTHP-1 cells to express genes and secrete interleukin (IL)-6 and monocyte chemoattractant protein (MCP)-1 proteins (Fig. 2a,b).

(a) TNF- gene expression and (b) TNF- production was examined in differentiated THP-1 (dTHP-1) cells stimulated with the supernatants from DP-1 and primary dental pulp cells (PriDPC)-1.

Amounts of TNF- in culture supernatants of original dental pulp cells were also measured (b: right).

Therefore, it can be concluded that this pro-inflammatory factor from dental pulp cells, DPIT, is a universal activator of monocytic cells.

(a) The production of IL-1, IL-6, and MCP-1, and (b) the gene expression of IL-6 and MCP-1 (b) in differentiated THP-1 (dTHP1) cells stimulated with the supernatants from DP-1 and primary dental pulp cells (PriDPC) were evaluated. dTHP-1 cells are stimulated with the supernatants from DP-1 (DP-1 sup) and PriDPC (PriDPC sup), LPS or Pam3CSK4 at indicated concentration, and total RNAs and supernatants are collected. Culture supernatants of dTHP-1 cells were collected 2 hrs after the stimulation and IL-1, IL-6, and MCP-1 protein levels were measured by ELISA.

Endogenous amounts of these cytokines in culture supernatants of dental pulp cells were also measured (a).

Taken together, we confirmed that culture supernatants from dental pulp cells contained a factor that could strongly induce pro-inflammatory reactions.

Interestingly, many of the commonly up-regulated genes were family members of interferon-inducible genes, suggesting that dental pulp cells were constantly exposed to weak stress stimuli.

It is very unlikely that PKR alone is transported extracellularly from dental pulp cells

to recipient macrophages.

We, therefore, hypothesized that PKR may be transported from dental pulp cells to macrophages via extracellular vesicles.

Pre-incubation of dTHP-1 cells with PKR inhibitors also suppressed TNF- α production in dTHP-1 cells following dental pulp cell-derived supernatant stimulation (Fig. 6b).

Culture supernatants from dental pulp cells, both DP-1 and PriDPC-1, markedly induced phosphorylation of p38 MAP kinase (p38), c-Jun N-terminal kinase (JNK), and nuclear factor (NF)- κ B in dTHP-1 cells (Fig. 6d).

that DPIT activity was mediated by PKR derived from dental pulp cells, and that the possible transport machinery was extracellular particles.

To test this possibility, we first separated the contents of cell culture supernatants from dental pulp cells using different pore-size filters and found that DPIT activity was completely lost in a fraction with a particle size below 220 nm (Fig. 7a).

The most abundant microvesicles from dental pulp cells isolated by ultra-centrifugation had a particle size of 230–250 nm, and round-shaped morphology of microvesicles was identified by transmission electron microscopy (TEM) analysis (Fig. 7b,c).

To confirm the activity of these vesicles, we compared the TNF- α -inducing activity of these particles with those of microvesicles derived from Hek293 cells (Hek293-MVs) and MCF7 cells (MCF7-MVs) isolated in a similar manner, and confirmed that only dental pulp cell-derived microvesicles (DP-1-MVs and priDPC-1-MVs) exhibited TNF- α inducing activity at both the gene and protein levels (Fig. 7d,e).

Further, we stimulated DP-1 cells with several concentrations of DP-1-MVs in fresh serum-free medium to clarify the autocrine effects of dental pulp cell-derived Figure 4.

We then labeled dental pulp DP-1-MVs with a fluorescent dye and observed their cellular uptake by dTHP-1 cells.

In fact, AHNAK protein was detected in the cytosol of dental pulp cells, but not Hek293 cells (Fig. 10a), and also in dental pulp tissue (Fig. 10b).

In addition, AHNAK was detected in culture supernatants from dental pulp cells (Fig. 10c).

We also investigated the presence of PKR in extracellular vesicles from both dental pulp cells and the breast cancer cells MCF-7, and confirmed its presence in cancer cell-derived microvesicles (Fig. 11c).

The possibility was that dental pulp cell-derived microvesicles contained activated or at least pre-activated PKR in their compartments without activation of target substrates inside the vesicles and that the active kinase was trans-introduced to the recipient monocyte/macrophage cell line.

(d) Culture supernatants from dental pulp cells (DP-1, priDPC-1), but not those from gingival fibroblasts, markedly activated JNK, p38, and NF- κ B in dTHP-1 cells.

Based on these experiments, the following mechanism is proposed. In dental pulp cells, a sustained weak stress signal is evoked, resulting in activation of a series of interferon-inducible genes and PKR.

Activated PKR continuously promotes the formation and aggregation of stress granules containing active PKR, and the dental pulp cells release stress granule aggregates together with AHNAK into microvesicles.

Methods

Human dental pulp cells and tissues were collected in compliance with the Hiroshima University ethical guidelines for epidemiological research.

The preparation of immortalized dental pulp cells (DP-1) cells was described previously².

Primary dental pulp (PriDPC) cells were isolated from outgrowth of dental pulp tissue explanted from teeth extracted for orthodontic reason or pericoronitis³¹.

PKR is transported from dental pulp cells via microvesicles.

(a) Size exclusion of MVs from the supernatants of dental pulp cells with a pore size below 220 nm completely abolished DPIT activity.

Uptake of dental pulp cell-derived MVs by recipient dTHP-1 cells.

Conditioned medium (total: 10 L) was collected from immortalized dental pulp cells (DP-1) .

mRNAs were isolated from sub-confluent immortalized dental pulp cells, primary dental pulp cells and control gingival fibroblasts.

(a c) Giant molecule AHNAK was detected in DP-1 and PriDPCs (a), dental pulp tissue (b), and from PriDPC supernatants (c).

Human dental pulp tissues dissected from healthy teeth extracted for orthodontic purposes were fixed with 3.7% formaldehyde for 24 h, sequentially immersed in 15% sucrose in PBS overnight and 30% sucrose overnight, embedded in OCT compound, and sectioned at 10- μ m thickness.

Rehydrated sections were treated as described above to detect AHNAK expression in human dental pulp tissues.

Acknowledgements

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Author Contributions

K.O. prepared immortalized human dental pulp cells.